TECHNICAL SPECIFICATIONS

SECTION 23 81 23

CRAH / COOLING UNITS

Project: Meridian Data Centre Campus, Phase 1
Navi Mumbai, India
Revision: 2024 (December 12, 2024)

PART 1 - GENERAL

1.1 Summary

This section specifies chilled water perimeter computer room air handlers (CRAH) with EC fans for server hall thermal management.

1.2 References

- AHRI 1360 - Performance rating of computer room air conditioners
- ASHRAE 90.1 - Energy standard for buildings
- IS 1391 - Room air conditioners

1.3 Submittal Requirements

A. Contractor shall submit the following for Engineer's approval prior to equipment procurement:

1. Product Data: Technical datasheets detailing all electrical and physical parameters, performance curves, and compliance.
2. Shop Drawings: Dimensional layouts showing clearance requirements, cabling paths, and weights.
3. Compliance Matrix: Clause-by-clause confirmation of conformance. Any deviation must be explicitly noted.
4. Factory Test Certificates and Type Test Certifications.

1.4 Quality Assurance

A. Equipment shall be manufactured by a firm specializing in the equipment class with a minimum of twenty (25) years of continuous commercial manufacturing history.

B. As specified in Section 26 05 00 Part 1.4.3.B, all medium voltage switchgear and high-power electrical components must be supplied with certified test certificates from an internationally recognized independent laboratory (ASTA, KEMA, or equivalent) confirming compliance with IEC standards.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

A. Subject to compliance with technical requirements, approved manufacturers are limited to the following:

2.2 Performance and Design Requirements

sensible_cooling: CRAH sensible cooling capacity must be >= 250 kW at 10°C entering chilled water temp.

airflow: Volumetric airflow rate must be >= 64,320 m3/h (approx. 37,850 CFM).

water_delta: Chilled water temperature drop (Entering to Leaving) must be 8 K.

efficiency: EC fan power density must be < 0.10 W/(m3/h) at design operating point.

refrigerant: Cooling coil circuit must utilize R-410A refrigerant.

Injected Contract Defects (Internal Quality Control Audit Targets):

- **DEF-004:** CRAH cooling unit COP/efficiency is specified as 'minimum COP of 5.2' in Part 2.2.4.C with no operating conditions (no entering/leaving water temperatures or airflow speed specified), making the performance claim unverifiable. (Location: 23 81 23 Part 2.2.4.C)

PART 3 - EXECUTION

3.1 Installation and Examination

A. Verify that structural concrete floor slab is fully cured. Verify structural load limit of 2,800 kg/m^2 is not exceeded in UPS or battery rooms.

B. Maintain a minimum clear work clearance of 1000 mm in front of all enclosures, and 1200 mm above, in accordance with local Indian Electricity Rules and IEC standards.

3.2 Field Quality Control and Site Commissioning

A. Startup services shall be performed by factory-authorized technicians.

B. Commissioning tests must be scheduled and carried out in accordance with the project's overall Cx Test Register. Every testing step must map back to a specific specification clause.

airflow_checks: Measure fan airflow using duct traverse traverse method.

capacity_runs: Conduct steady-state heat load run to verify sensible cooling capacity.